

Radio Access Technologies

1) **Orthogonal Frequency Division Multiplexing (OFDM):**

Orthogonal Frequency Division Multiplexing (OFDM) is the modulation technique used in 5G to transmit data efficiently over multiple subcarriers. Instead of sending all data through one carrier frequency, OFDM divides the available bandwidth into many smaller orthogonal subcarriers that transmit data simultaneously.

Advantages:

- High data rates
- Better spectrum efficiency
- Reduced interference
- Improved resistance to multipath fading
- Supports multiple users simultaneously

OFDM Structure:

Total Channel Bandwidth

| SC1 | SC2 | SC3 | SC4 | SC5 | SC6 | . . . | SCn |

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Each Subcarrier Transmits Data Independently

- 2) **Subcarriers:** A subcarrier is a small frequency channel within the total available bandwidth. Each subcarrier carries a portion of the user's data, allowing parallel transmission and improving efficiency.

Advantages:

- Parallel data transmission
- High throughput
- Reduced interference
- Better reliability

3) **Numerology:** Numerology defines the subcarrier spacing (SCS) used in 5G NR. Different applications require different spacing depending on latency, coverage, and frequency.

Subcarrier Spacing	Typical Usage
15 kHz	Wide coverage (FR1)
30 kHz	General 5G deployment
60 kHz	Low latency applications
120 kHz	Millimeter Wave (FR2)
240 kHz	High-frequency applications

Relationship:

Higher Subcarrier Spacing

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Lower Symbol Duration

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Lower Latency

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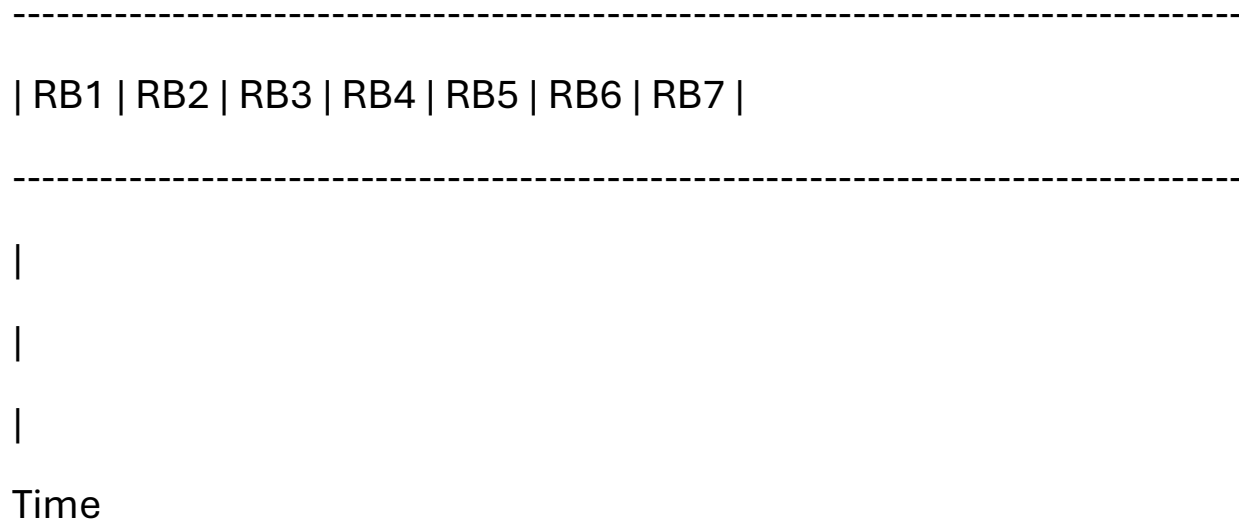
Higher Data Rate

4) **Resource Blocks (RB):** A Resource Block (RB) is the smallest unit of radio resources allocated to a user in a 5G network.

One Resource Block contains:

- 12 consecutive subcarriers
- Multiple OFDM symbols over time

Resource Block Diagram



Purpose:

- Allocates bandwidth to users
- Efficient radio resource scheduling
- Supports multiple users simultaneously

5) **Beamforming:** Beamforming is an antenna technology that focuses radio signals directly toward a user instead of broadcasting equally in all directions

Benefits:

- Stronger signal
- Higher throughput
- Better coverage

- Reduced interference
- Increased network capacity

6) **Analog Beamforming:** In Analog Beamforming, all antenna elements share a single RF chain. The phase of the transmitted signal is adjusted to steer the beam.

Advantages:

- Lower cost
- Lower power consumption
- Simpler implementation

Limitations:

- Supports only one beam at a time
- Limited flexibility

7) **Digital Beamforming:** Digital Beamforming provides each antenna with its own RF chain and digital signal processing.

Advantages:

- Multiple beams simultaneously
- Better interference management
- Higher accuracy
- Supports multiple users

8) **Massive MIMO:** Massive Multiple Input Multiple Output (Massive MIMO) uses many transmitting and receiving antennas to improve spectral efficiency, throughput, and network capacity.

Advantages:

- Higher throughput

- Better coverage
- Improved reliability
- Increased capacity
- Supports many users simultaneously

A) **4T4R**: 4 transmit antennas and 4 receive antennas

Applications:

- Small cells
- Indoor deployments
- Rural coverage

B) **32T32R**: 32 transmit antennas and 32 receive antennas

Applications:

- Urban environments
- Medium-capacity networks

C) **64T64R**: 64 transmit antennas and 64 receive antennas

Applications:

- High-density urban areas
- Stadiums
- Airports
- Smart cities

Provides:

- Maximum throughput
- Excellent beamforming
- Large user capacity
- High spectral efficiency

9) **5G Spectrum:** The 5G spectrum is divided into two frequency ranges.

A) FR1 (Frequency Range 1):

Frequency: 410 MHz-7.125 GHz also called: Sub-6 GHz

Characteristics:

- Long coverage
- Better indoor penetration
- Moderate speed
- Lower attenuation

Applications:

- Nationwide coverage
- Smart cities
- Rural networks
- General mobile broadband

B) FR2 (Frequency Range 2):

Frequency: 24.25 GHz-71GHz also called: Millimeter Wave (mmWave)

Characteristics:

- Extremely high speed
- Ultra-low latency
- Very high bandwidth
- Short coverage
- Easily blocked by obstacles

Applications:

- Stadiums
- Airports
- Industrial automation
- AR/VR
- Smart factories